Supplementary material

Subject-independent surface EMG electrode reduction using cross-fitted attribution: a prespecified non-inferiority study

S1 Departures from the analysis plan

The following changes were made after the statistical analysis plan was written. None altered a prespecified decision rule (margin, statistical unit, bootstrap or definition of k^*).

1. The plan required Holm correction within H1. It was applied after the primary analysis and left k^* unchanged at all three margins (table S1). A fixed-sequence analysis, testing $k = 15, 14, \ldots$ and stopping at the first failure, is reported in the same table as a sensitivity analysis.
2. Training-sample sizes (40,000 windows for the support vector machine, 200,000 for the random forest and for the electrode-selection analyses) were chosen from held-out Macro-F1 on two folds (participants 1 and 7) before the main analyses. For these two participants the held-out data therefore informed a design choice.
3. The plan did not specify a criterion for choosing λ, so none was chosen and all four values are reported. The value $\lambda = 0.5$ used for SR-CFA in the H2 comparison was not chosen by a stated criterion, so the $\lambda = 0$ ranking is reported alongside it.
4. The detailed H2–H4 test families (tests and subset sizes) were specified after the performance-curve and selection-method results were available.
5. The comparison model for H2 and H4 was reduced from 150 to 50 trees for computational cost before any comparison result was available.
6. An initial count-feature threshold 87 times smaller than the final GRABMyo value saturated WAMP and was replaced. The leave-one-out check of the thresholds (section S2) was added after the primary analysis.
7. An initial deletion analysis combined deletion orders computed with different training sizes; it was replaced by the analysis reported here, in which all orders share one configuration (section S3).
8. The bipolar analysis was run three times. The first run used the monopolar count-feature threshold; the second used 5×10^{-3} mV, whose stated derivation could not be reproduced; the reported analysis uses 4×10^{-3} mV, the value given by the documented rule. Earlier outputs are archived.
9. The fixed-layout and feature-redundancy analyses (sections S4 and S5) were not in the plan.
10. The window-length sensitivity analysis and the 17-class task, listed as exploratory in the plan, were not run.

S2 Leave-one-out check of the count-feature thresholds

All thresholds follow one rule: 1.15 times the median absolute first difference of the signal, rounded to one significant figure. The GRABMyo threshold was derived from 54 unfiltered recordings of 6 participants. For every outer fold we recomputed the median without the held-out participant and applied the same rule. It gave 10^{-2} mV in 43 of 43 folds (leave-one-out medians 8.52–9.45×10^{-3} mV); only 6 folds held out a participant who contributed to the sample. For the derived bipolar signal (median 3.40×10^{-3} mV on the same recordings) it gave 4×10^{-3} mV in 43 of 43 folds. For DB5 it gave 7 quantisation steps in 10 of 10 folds. Deriving the thresholds within each fold would therefore have produced identical features. Because the thresholds were not estimated within folds, we describe this as a verification rather than as fold-local estimation.

S3 Consistency of the deletion analysis

The Remove-Bottom subsets coincide with the H2 cross-fitted SHAP subsets in all 172 participant-by-size cells, and their Macro-F1 values agree to within a mean absolute difference of 0.0014 (maximum 0.0084). The small differences arise because features were standardised separately for each subset in the deletion analysis and once over all features in the H2 comparison.

S4 Fixed layout

A single 12-electrode layout was derived from a cross-fitted ranking ($\lambda = 0.5$) on the 22 odd-numbered participants, removing electrodes 3, 9, 11, 12. Models with 16 and with 12 electrodes were trained on those participants and tested on the 21 even-numbered participants. The 12-electrode model scored −0.0124 against the 16-electrode model (95% confidence interval −0.0179 to −0.0077), within the margin. The electrode count was chosen after inspecting the full-cohort performance curve, which included the even-numbered participants. Although electrode identities were selected using development participants only, this is therefore not an independent confirmation of the chosen count. It is a single-split consistency check.

S5 Feature redundancy

Integrated EMG equals 512 times the mean absolute value for these windows. Repeating the comparison-model evaluation without integrated EMG changed Macro-F1 by −0.0005 (95% confidence interval −0.0021 to +0.0010; 43 folds). The median per-fold Spearman ρ between rankings with and without integrated EMG was 0.994, and the top-12 set was identical in 34 of 43 folds, so the duplicated feature had little measurable effect on the ranking.

S6 Participant-level attribution extension

The post-hoc extension uses the same out-of-fold attributions as the original ranking, but gives each outer-training participant equal weight before computing the median and median absolute deviation. All penalties and subset sizes were evaluated with the same training budget as the original curve. Every reused fit records its source, and inputs are fingerprinted to prevent mixing incompatible runs. Between-person dispersion is descriptive heterogeneity, not a standard error of attribution. It includes both participant differences and differences between the inner fitted models.

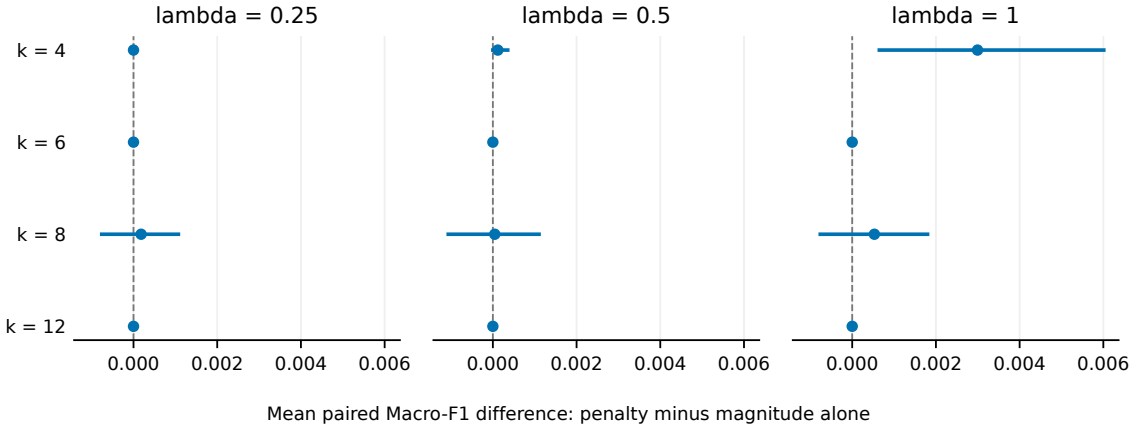


Figure S1. Does the upgraded stability penalty improve prediction? Each point compares a positive penalty with magnitude alone under participant-level aggregation. Bars are unadjusted participant-bootstrap 95% intervals; asterisks indicate Holm-adjusted $p < 0.05$ across all penalty-by-size comparisons. Positive differences favour the penalty. Zero-width intervals arise when all paired results coincide. This diagnostic is exploratory and does not replace the original H3 test.

S7 Supplementary tables

Table S1. H1 with Holm correction over the 15 subset sizes (43 participants, $\lambda = 0$, $\delta = 0.02$). p = one-sided cluster-bootstrap p for $H_0 : \bar{\Delta}_k \leq -\delta$ (10,000 resamples); Holm over the 15 tested k at $\alpha = 0.025$. Fixed sequence (sensitivity analysis): $k = 15, 14, \ldots$ tested in order, stopping at the first failure.

k	$\bar{\Delta}_k$	Lower bound	p	p_{Holm}	Non-inf.	Non-inf. (Holm)	Fixed sequence
15	−0.0036	−0.0056	<0.001	0.002	yes	yes	yes
14	−0.0060	−0.0081	<0.001	0.002	yes	yes	yes
13	−0.0109	−0.0133	<0.001	0.002	yes	yes	yes
12	−0.0133	−0.0160	<0.001	0.002	yes	yes	yes
11	−0.0204	−0.0246	0.579	1.000	no	no	no
10	−0.0259	−0.0311	0.984	1.000	no	no	no
9	−0.0318	−0.0387	0.999	1.000	no	no	no
8	−0.0366	−0.0440	1.000	1.000	no	no	no
7	−0.0413	−0.0489	1.000	1.000	no	no	no
6	−0.0610	−0.0713	1.000	1.000	no	no	no
5	−0.1133	−0.1277	1.000	1.000	no	no	no
4	−0.1444	−0.1590	1.000	1.000	no	no	no
3	−0.2015	−0.2212	1.000	1.000	no	no	no
2	−0.2420	−0.2603	1.000	1.000	no	no	no
1	−0.3603	−0.3832	1.000	1.000	no	no	no

k^* (unadjusted / Holm / fixed sequence) at $\lambda = 0$: $\delta = 0.01$: 14 / 14 / 14; $\delta = 0.02$: 12 / 12 / 12; $\delta = 0.03$: 11 / 11 / 11; identical for all four λ.

Table S2. Stability regularisation versus attribution magnitude alone on GRABMyo (H3): $Q(\lambda)$ ranking minus $Q(0)$ ranking on the sparse curve. Tied = participants for whom both rankings select the identical subset. Holm over 12 tests.

λ	k	Mean diff	95% CI	Tied	p_{Holm}
0.25	4	−0.0002	−0.0005 to +0.0000	41/43	1.000
0.25	6	+0.0000	+0.0000 to +0.0000	43/43	1.000
0.25	8	+0.0001	+0.0000 to +0.0002	42/43	1.000
0.25	12	+0.0000	+0.0000 to +0.0000	43/43	1.000
0.5	4	+0.0004	−0.0005 to +0.0018	40/43	1.000
0.5	6	+0.0000	+0.0000 to +0.0000	43/43	1.000
0.5	8	+0.0008	+0.0000 to +0.0019	40/43	1.000
0.5	12	+0.0000	+0.0000 to +0.0000	43/43	1.000
1	4	+0.0010	−0.0003 to +0.0027	38/43	1.000
1	6	+0.0000	+0.0000 to +0.0000	43/43	1.000
1	8	+0.0015	+0.0004 to +0.0028	36/43	0.216
1	12	−0.0005	−0.0014 to +0.0000	42/43	1.000

Table S3. Participant-level attribution extension (exploratory; GRABMyo). Non-inferiority uses the same-budget full-array reference and the original engineering margin. The paired difference compares participant versus fold aggregation at the same penalty and electrode count.

λ	k^*	k^* (Holm)	Δ at $k = 12$ (95% CI)	p_{Holm}
0	12	12	+0.0000 (+0.0000, +0.0000)	1.000
0.25	12	12	+0.0000 (+0.0000, +0.0000)	1.000
0.5	12	12	+0.0000 (+0.0000, +0.0000)	1.000
1	12	12	+0.0005 (+0.0000, +0.0014)	1.000

Table S4. NinaPro DB5 channel-count performance curve (exploratory; cross-fitted ranking, $\lambda = 0$; 10 subjects; reference $k = 16$, mean Macro-F1 0.281; margin $\delta = 0.02$). Lower 95% bound shown to six decimals because the $k = 12$ decision lies at the margin. Non-inferior: lower bound above $-\delta$ (uncorrected), and after Holm correction over 15 subset sizes; bootstrap seeds: share of 200 alternative seeds under which k is non-inferior.

k	Macro-F1	Δ_k	Lower bound	Non-inf.	Non-inf. (Holm)	Bootstrap seeds
15	0.270	−0.0108	−0.023773	no	no	0%
14	0.269	−0.0119	−0.027438	no	no	0%
13	0.266	−0.0143	−0.031186	no	no	0%
12	0.268	−0.0126	−0.019994	yes	no	96%
11	0.260	−0.0211	−0.034121	no	no	0%
10	0.265	−0.0160	−0.031368	no	no	0%
9	0.260	−0.0202	−0.036943	no	no	0%
8	0.252	−0.0288	−0.055369	no	no	0%
7	0.232	−0.0486	−0.075346	no	no	0%
6	0.231	−0.0498	−0.081816	no	no	0%
5	0.200	−0.0803	−0.118422	no	no	0%
4	0.180	−0.1006	−0.150266	no	no	0%
3	0.162	−0.1185	−0.174145	no	no	0%
2	0.138	−0.1430	−0.195307	no	no	0%
1	0.105	−0.1754	−0.229548	no	no	0%

Table S5. GRABMyo derived bipolar pairs $F_j - F_{j+8}$ (exploratory; 43 participants; $\lambda = 0$; count-feature threshold 4×10^{-3} mV). Reference = $k = 8$ model of the same curve (200,000 training windows), mean Macro-F1 0.480; margin $\delta = 0.02$; bootstrap seeds: share of 200 alternative seeds under which k is non-inferior.

k	Macro-F1	Δ_k	95% CI	Non-inf.	Bootstrap seeds
7	0.464	−0.0158	−0.0203 to −0.0113	no	0%
6	0.442	−0.0375	−0.0451 to −0.0303	no	0%
5	0.405	−0.0742	−0.0845 to −0.0646	no	0%
4	0.356	−0.1240	−0.1392 to −0.1096	no	0%
3	0.310	−0.1697	−0.1852 to −0.1546	no	0%
2	0.240	−0.2399	−0.2615 to −0.2177	no	0%
1	0.174	−0.3052	−0.3298 to −0.2790	no	0%

Table S6. Cross-session evaluation (exploratory). All conditions are also subject-independent: the test participant is never in training.

Design	Train sessions	Test session	Train windows	Macro-F1
CS1	1	1	183,456	0.492 ± 0.101
CS1	1	2	183,456	0.528 ± 0.088
CS1	1	3	183,456	0.510 ± 0.090
CS2	1+2	3	366,912	0.530 ± 0.095

Paired contrasts (exploratory, uncorrected): CS1: test S2 − test S1: +0.0355 (95% CI +0.0091 to +0.0621), p = 0.024; CS1: test S3 − test S1: +0.0172 (95% CI −0.0038 to +0.0391), p = 0.272; S3 test: CS2 (train S1+S2) − CS1 (train S1): +0.0210 (95% CI +0.0146 to +0.0274), $p < 0.001$.

Table S7. Full-array XGBoost performance by active GRABMyo gesture. Means and percentile confidence intervals are computed across participants. Gesture identifiers follow the public dataset.

Gesture	Mean F1	95% participant interval
1	0.444	0.389 to 0.496
2	0.309	0.256 to 0.360
3	0.526	0.472 to 0.577
4	0.349	0.297 to 0.399
5	0.486	0.424 to 0.547
6	0.602	0.532 to 0.667
7	0.506	0.443 to 0.566
8	0.499	0.441 to 0.553
9	0.388	0.333 to 0.443
10	0.538	0.470 to 0.603
11	0.942	0.927 to 0.956
12	0.828	0.775 to 0.874
13	0.814	0.770 to 0.852
14	0.491	0.431 to 0.550
15	0.545	0.489 to 0.598
16	0.410	0.368 to 0.449